CS-4982 Progress Report-2

Darin Thomson; 3776723; UNBSJ

1st March, 2026

1 Updated Technical Project Description (Changes):

The project has been on-schedule according to the timeline/ milestone table adopted during the drafting of the proposal and the progress report-1. This includes the brain-storming of the topic/ discussion with my co-author on topics related to this research and exchanging information and key resources related to and surrounding the topic. The drafting of the proposal had taken a little more time than expected. This increased the duration of drafting the proposal from 2 days (Jan 23rd - Jan 25th) to 3 days (Jan 22nd - Jan 25th).

The research process had begun as soon as the draft of the proposal was done and submitted. This process took longer than the allotted time and had gone over until Feb 5th instead. This caused the drafting of progress report - 1 to be moved to Feb 6th-8th. This had caused a delay on the first draft of the essay and the drafting of progress report - 2 (Feb 9th - March 1st - March 22nd) slightly further than schedule. The remaining of the tasks should follow the timeline with the possibility of more tweaks/ modifications. The time after the submission of progress report-1 (Feb 9th - March 1st) had mostly been consumed for the drafting of the technical report and summarizing the entire process until now into the progress report-2. Apart from that, I spent substantial amount of this time (until now) on the implementation/ code aspect of my deliverable. A small component of my time was also spent recording the video required for the submission of progress report-2.

The implementation included running a python code simulation on google's colab environment. This shows a realistic demonstration of my study, taking input and producing the desired output using a dashboard. The implementation is a python simulation that walks us through the full pipeline using EDGAR (an open-source python library), as well as some synthetic sample data. It chiefly covers 5 key stages including: 1) data collection/ input 2) data cleaning and preprocessing 3) Feature correlation/ extraction 4) hedging index construction and 5) predictive modeling and Validation.

The most challenging parts of the analysis and code simulation were the feature correlation and the creation of dashboards to display the models' findings. More details on the implementation can be found in the video!

2 Revised Summary of Target Deliverables:

2.1 Researching Process

2.2 Draft Proposal

2.3 Draft of progress report-1

2.4 Start drafting the first/ initial draft of the technical report

2.5 Draft of progress report-2

2.6 Video recording as part of progress report-2

2.7 Explore other possible resources related and surrounding the research. This would possibly involve a collaboration with my co-author. Note: This has not been included as part of my updated GANTT chart as it will be a part of my initial draft process

2.8 Essay final draft which would involve a strong refinement of the initial draft of the essay

2.9 Lastly, the final commitment to accomplish would be the submission of the report including all the required deliverables (as mentioned above) and successfully presenting my research report to Dr. Baker and my peers.

3 Mature Solution Architect Diagram:

4 Updated GANTT Chart/ Timeline:

5 Updated Unforeseen Challenges / Unknown Issues:

A key challenge faced until the drafting of progress report-2 was that the operational/ implementation model such as the code and software aspects of the project had taken lot more time and study than anticipated. However, this did not cause great changes to the timeline or the GANTT chart as it had been accommodated within the drafting of the initial report and the progress report-2 timeline.

Apart from that, the biggest unforseen challenge faced was during the creation of the simulation, especially in the feature extraction segment and the

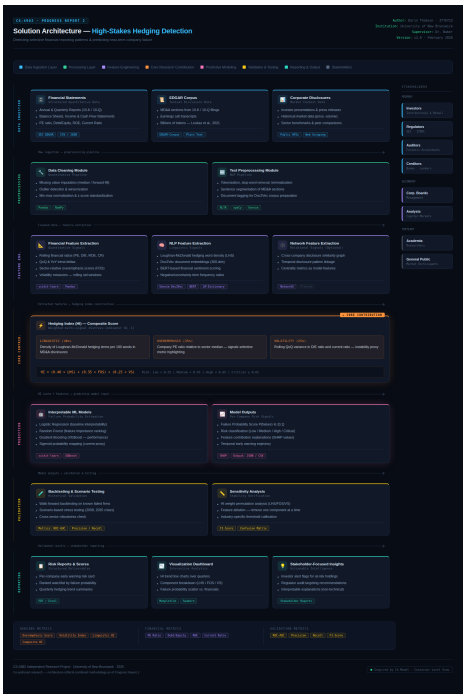

Figure 1: Updated Solution Architecture Diagram

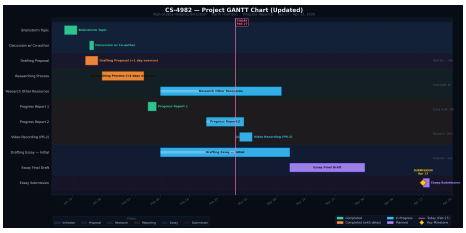

Figure 2: Updated GANTT Chart

modeling/ dashboard generation segment. That aside, everything has been going smoothly with an efficient implementation of each of the steps of the process.

6 Updated Project Evaluation Tasks and Outline:

The tasks required to be performed would remain the same as mentioned in the initial proposal draft and the progress report-1. The current evaluation metrics:

6.1 How to test the prototype?

The prototype, when ready, could be tested using backtesting strategies, scenario based testing, and potentially using sensitivity analysis.

6.2 Success criteria of the Project

(a) If an accurate prediction of near-term company distress or failure is made using the research.

(b) Being able to identify statistically significant hedging behaviors of the companies considered.

(c) Robustness of our research/ approach across industries and time periods.

(d) Interpretable results that can potentially be used by our stakeholders as mentioned.

6.3 Metrics adopted

Hedging metrics (like overemphasis, volatility, and hedging index), financial metrics (including PE ratios, debt-equity ratios, earnings ratios, etc.) and a few other validation metrics.

7 Updated References:

The references and key resources used during the process of this research study are as follows. As of now, there may still be potential additions/ subtractions by the end of this study.

7.1 References that describe the problem statement -

1. Loughran, T., and McDonald, B. (2024). Loughran-McDonald Master Dictionary with SentimentWord Lists—Useful for explaining how corporate language can signal hedging or selective disclosure.

2. Loukas, L., Fergadiotis, M., Androutsopoulos, I., and Malakasiotis, P. (2021). EDGAR-Corpus: Billions of Tokens Make The World Go Round. ECONLP. — Provides context for large-scale analysis of corporate filings and textual disclosures.

3. Abney, S. (2007). Semi-Supervised Learning for Computational Linguistics. CRC Press. —Supports understanding the role of NLP in detecting hidden signals in text.

7.2 References that illustrate the Approach/ Methodology Adopted -

4. Rehurek, R. (Gensim). Doc2Vec Tutorial — Shows how document embeddings can convert textual disclosures into numerical features for modeling.

5. Loukas et al., EDGAR-Corpus — Also supports methodology by providing structured large-scale data for model training.

6. Enhancing Financial Sentiment Analysis with BERT - Example of advanced NLP methods for predictive modeling of corporate disclosures.

7.3 References that report on the resources adopted -

7. NetworkX Developers. NetworkX Documentation — For network-based feature analysis if needed.

8. Gensim / Doc2Vec Tutorial — For NLP and embeddings (also part of methodology, but primarily a resource/tool).

9. Loughran-McDonald Master Dictionary — Used as a lexical resource for sentiment/hedging detection.

10. EDGAR-Corpus dataset — Provides the textual data used as part of the process.

11. Python libraries like Pandas, NumPy, scikit-learn, matplotlib.

12. Abney, S. (2007) — Provides methodology guidance for NLP tools you will use.

8 Video demonstration (attached):